

Capstone 3 Report: Dashboard Development

European Urban Tree Cover Analysis

Deliverables Overview

Step	Deliverable	File
Step 1	Dashboard Plan	dashboard_plan.md
Step 2	Database Connection & SQL Queries	sql/queries.sql , sql/db_connection.py
Step 3	N/A	—
Step 4–5	Dashboard Code & Screenshots	app.py , templates/*.html , static/style.css
Step 6	UX Notes & Testing	screenshots/ux_notes.md

Step 1: Dashboard Plan

Case Study

How does urban tree canopy cover relate to population density across European cities? This project analyzes the relationship between population density, tree canopy cover, and tree equity across 98 cities in 42 European countries.

Equity Categorization

Cities are classified into four equity tiers based on quartile breaks of the population-to-tree ratio (people per km² of tree cover):

Category	Ratio Range	Cities
Excellent	<50,844	25
Good	50,844–134,561	25
Fair	134,561–236,415	24
Needs Improvement	>236,415	24

Dashboard Types

Dashboard	Type	Purpose	Route
European Tree Cover Atlas	Strategic	High-level geographic overview and monitoring	/dashboard1
City Tree Equity Analyzer	Analytical	Deep-dive comparison and exploration	/dashboard2

Visualization Techniques

Quantitative (6 total, ≥4 required):

- Bar chart (D1 — top 10 tree cover %)
- Scatter plot (D1, D2 — population vs tree %, density vs tree density)
- Donut chart (D1 — equity category distribution)
- Horizontal bar chart (D2 — pop-to-tree ratio ranking)
- Grouped bar chart (D2 — city comparison across 4 metrics)
- KPI metric cards (D2 — numerical display)

Qualitative (5 total, ≥3 required):

- Geospatial map (D1 — Leaflet map with equity-colored markers)
- Sortable data tables (D1, D2 — full city listings)
- KPI metric cards (D2 — at-a-glance stats)
- Color-coded equity categories (D1, D2 — consistent green/yellow/red)
- Interactive filters (D1, D2 — dropdowns, sliders, checkboxes, search)

Step 2: Database Integration

Database Source

The database from Capstone 2 (`capstone_1/data/kemv.db`) was expanded with a new `city_tree_rankings` table containing 98 cities across 42 countries. This table combines data from the original `cities` , `countries` , `urban_trees` , and `city_rankings` tables into a single flat structure optimized for dashboard queries.

Database Connection

File: `sql/db_connection.py`

- SQLite connection with row factory for dictionary access
- PRAGMA optimizations: `journal_mode=WAL` , `cache_size=-8000` (64 MB)
- 7 query functions for dashboard data retrieval

SQL Queries

File: `sql/queries.sql`

Query	Purpose	Tables Joined
All Cities with Full Metrics	Map, tables, charts	4 (cities + countries + urban_trees + city_rankings)
Top 10 by Tree Cover %	D1 bar chart	3
Equity Category Distribution	D1 donut chart	1 (city_rankings)
Countries List	D1 filter dropdown	2
Filtered Cities by Country	D1 country filter	4
Single City Detail	D2 KPI cards	4
Filtered by Population + Equity	D2 tables/charts	4

The `db_connection.py` queries an additional `city_tree_rankings` table (not present in the Capstone 2 schema) for the main `/api/cities` endpoint, while the remaining functions query the original normalized tables.

Step 4–5: Dashboard Implementation

Tech Stack

- **Backend:** Flask (Python) + sqlite3
- **Frontend:** HTML5, CSS3, Chart.js 4.4.7 (CDN), Plotly.js (CDN), Leaflet 1.9.4 (CDN)
- **Database:** SQLite 3

Application Structure

```

capstone_3/
├── app.py                # Flask application (routes + API endpoints)
├── dashboard_plan.md     # Step 1 deliverable
├── sql/
│   ├── db_connection.py  # Database connection and query functions
│   └── queries.sql       # SQL query reference
├── templates/
│   ├── dashboard1.html   # European Tree Cover Atlas
│   └── dashboard2.html   # City Tree Equity Analyzer
├── static/
│   └── style.css         # Global styles
└── screenshots/
    ├── dashboard1.png    # D1 screenshot
    ├── dashboard2.png    # D2 screenshot
    └── ux_notes.md       # Step 6 UX documentation

```

API Endpoints

Endpoint	Method	Returns
/	GET	Redirects to Dashboard 1
/dashboard1	GET	Dashboard 1 HTML
/dashboard2	GET	Dashboard 2 HTML
/api/cities	GET	All 98 cities with metrics
/api/top10	GET	Top 10 by tree cover %
/api/equity-distribution	GET	Counts per equity category
/api/countries	GET	Country list for filters
/api/city-detail?name=	GET	Single city detail
/api/filtered-cities?...	GET	Filtered city list

Dashboard 1: European Tree Cover Atlas (Strategic)

Target Users: Urban planners, policy makers, environmental advocates

Visualizations:

1. **Leaflet map** — City markers colored by equity category, sized by population, with click popups
2. **Summary stat cards** — City count, avg tree cover %, best/worst city
3. **Donut chart** — Equity category distribution (Chart.js)
4. **Bar chart** — Top 10 cities by tree cover % (horizontal, Chart.js)
5. **Scatter plot** — Population vs tree cover %, colored by equity (Chart.js)
6. **Box plot** — Population-to-tree ratio distribution with quartile bands (Plotly)
7. **Sortable data table** — All cities with rank, tree %, density, ratio

Interactive Elements:

- Country dropdown filter
- Equity category dropdown filter
- City search text input

Dashboard 2: City Tree Equity Analyzer (Analytical)

Target Users: Researchers, data analysts, journalists

Visualizations:

1. **KPI cards** — Selected city name, tree %, equity rank, population density
2. **Horizontal bar chart** — All cities ranked by pop-to-tree ratio (Chart.js)
3. **Scatter plot** — Population density vs tree density (Chart.js)
4. **Grouped bar chart** — Compare 1–2 cities across 4 metrics (normalized %, Chart.js)
5. **Sortable equity ranking table** — Full dataset with color-coded rows

Interactive Elements:

- Multi-select city dropdown (Ctrl+click)
 - Population range slider (dual input)
 - Equity category checkboxes
 - Sortable table columns
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Step 6: User Experience & Testing

UX Improvements

1. **Responsive layout** — CSS Grid/Flexbox; 2-column desktop, single-column mobile
2. **Consistent color coding** — Green/yellow/red equity categories across all charts, map, and tables
3. **Equity badges** — Pill-shaped labels for quick scanning in tables
4. **Tooltip information** — All charts show city name, country, and metrics on hover
5. **Single API call** — Dashboard 1 loads all data once and filters client-side
6. **Chart reuse** — Destroys/recreates chart instances to prevent memory leaks

Key Performance Optimizations

- SQLite WAL mode and 64 MB cache for faster queries
 - Client-side filtering for instant responsiveness
 - Single API call for Dashboard 1 (98 cities loaded once)
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Key Findings

Metric	Value
Cities analyzed	98
Countries represented	42
Average tree cover	3.27%
Best tree cover	Vatican City (16.90%)
Worst tree cover	Reggio Calabria (0.01%)
Largest city	London (8.98M)
Smallest city	Vatican City (800)
Dataset expanded from	30 cities (Capstone 2) → 98 cities (Capstone 3)

Top 10 Cities by Tree Cover %

Rank	City	Country	Tree %
1	Vatican City	Vatican City	16.90%
2	San Marino	San Marino	8.25%
3	Bern	Switzerland	7.08%
4	Reykjavik	Iceland	6.77%
5	Dublin	Ireland	6.74%
6	Merida	Spain	6.70%
7	Tallinn	Estonia	6.68%
8	Bremen	Germany	6.55%
9	Riga	Latvia	6.54%
10	Leipzig	Germany	6.52%

Files

File	Description
app.py	Flask application with 8 API endpoints
templates/dashboard1.html	Dashboard 1: European Tree Cover Atlas
templates/dashboard2.html	Dashboard 2: City Tree Equity Analyzer
static/style.css	Global styles and responsive layout
sql/db_connection.py	SQLite connection and 7 query functions
sql/queries.sql	Reference SQL queries
dashboard_plan.md	Step 1 — Dashboard plan and design
screenshots/ux_notes.md	Step 6 — UX improvements and testing